

Chapter 5: Sets, Sequences & Matrices

Set Operations, AP/GP Progressions, Matrix Algebra & Determinants

5.1 Set Operations & De Morgan's Laws

Sets are collections of distinct objects. Key set operations include Union ($\cup$), Intersection ($\cap$), Difference ($\setminus$), and Complement (A').

De Morgan's Laws

1. $(A \cup B)' = A' \cap B'$

2. $(A \cap B)' = A' \cup B'$

Cardinality Formula: $|A \cup B| = |A| + |B| - |A \cap B|$

5.2 Arithmetic & Geometric Sequences

Arithmetic Progression (AP)

Common Difference: d

n -th Term: $a_n = a_1 + (n - 1)d$

Sum of n terms:

$$S_n = (n/2)[2a_1 + (n - 1)d] = (n/2)(a_1 + a_n)$$

Geometric Progression (GP)

Common Ratio: r

n -th Term: $a_n = a_1 \cdot r^{n-1}$

Sum of n terms: $S_n = a_1(1 - r^n) / (1 - r)$

Infinite Sum ($|r| < 1$): $S_{\infty} = a_1 / (1 - r)$

Worked Example 5.1: Infinite Geometric Series

Problem: Find the sum of the infinite series: $12 + 4 + 4/3 + 4/9 + \dots$

Solution: $a_1 = 12$, common ratio $r = 4/12 = 1/3$.

Since $|1/3| < 1$, the infinite sum converges:

$$S_{\infty} = 12 / (1 - 1/3) = 12 / (2/3) = 18.$$

5.3 Matrix Operations & Determinants

A matrix is a rectangular array of numbers. Matrix multiplication $A \cdot B$ is defined only when the number of columns in A equals the number of rows in B .

2x2 Matrix Inversion

For $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$:

$$\det(A) = ad - bc$$

If $\det(A) \neq 0$, then:

$$A^{-1} = (1 / \det(A)) \cdot \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Cramer's Rule

For $AX = B$:

$$x = \det(A_x) / \det(A)$$

$$y = \det(A_y) / \det(A)$$

(Valid provided $\det(A) \neq 0$)

Worked Example 5.2: Matrix Inverse & Product

Problem: Find the determinant and inverse of matrix $A = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}$.

Solution:

$$\det(A) = (4)(3) - (2)(1) = 12 - 2 = 10.$$

$$A^{-1} = (1/10) \cdot \begin{bmatrix} 3 & -2 \\ -1 & 4 \end{bmatrix} = \begin{bmatrix} 0.3 & -0.2 \\ -0.1 & 0.4 \end{bmatrix}.$$

Chapter 5 Review & Practice Problems

- Given sets $U = \{1, 2, \dots, 10\}$, $A = \{2, 4, 6, 8\}$, $B = \{4, 5, 6, 7\}$, verify De Morgan's First Law.
- The 3rd term of an AP is 10 and the 8th term is 25. Find the 1st term and the sum of the first 15 terms.
- Evaluate the matrix product $A \cdot B$ where $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 \\ 1 & 5 \end{bmatrix}$.
- Solve the linear system using Cramer's Rule:

$$\begin{aligned} 3x + 4y &= 10 \\ 2x - y &= 3 \end{aligned}$$

